(HR of 0.91 [0.85-0.97], p = 0.006) were protective against the development of RR, while an increase in the apical height of the tumor (HR of 1.18 [1.06-1.31], p = 0.003) was detrimental. Increases in age (HR of 1.01[1.00-1.03], p = 0.05), increases in apical height (HR of 1.35 [1.22-1.48], $p<0.001$), and a greater total dose to the fovea (HR of 1.01 [1.01-1.01], $p<0.001$) were predictive of mild or severe vision loss.

Conclusions: Although plaque radiotherapy offers excellent tumor control (97%), almost one third of patients are left with visual acuity of ≤ 20/200. Modifiable factors associated with poorer visual outcomes such as total doses to the fovea and optic disc could be reduced by the use of alternative isotopes and improvements in plaque design.

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2174 Finger's Slotted Plaque Brachytherapy for Peripapillary, Juxtapapillary and Circumpapillary Choroidal Melanoma

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Purpose/Objective(s): There exists an unacceptable rate of failure of local control (12-25%) among patients treated with plaque irradiation for choroidal melanomas near, touching or surrounding the optic disc. This study reports 5-year results of patients treated with custom slotted eye plaques designed to improve localization for these anatomically challenged patients.

Materials/Methods: A single-center study was performed with 24 consecutive patients, treated between 1995 and January 2010. Recorded characteristics were related to the patient, tumor, ultrasonography, and outcomes. Follow-up data included change in visual acuity, tumor-size, recurrence, eye-retention and metastasis. Custom slotted plaques were fashioned from COMS-type shells as to include 8-mm wide, variable depth slots. Prescription doses ranged from 69.3-163.8 Gy (mean 85.0 Gy) based on delivering a minimum 85 Gy tumor dose to the tumor within the slot. Brachytherapy was continuously delivered over 5-7 day durations.

Results: At presentation, the mean patient age was 55 years (median: 53, range: 22-83 years). The choroidal melanomas were within 1.0 mm of the optic nerve in 12.5% (n = 3), juxtapapillary (touching up to 180 degrees of the optic disc) in 21% (n = 5), greater or equal to 180 degress in 25% (n = 6), circumpapillary (surrounding 360 degrees of the optic disc) in 37.5% (n = 9), and there was no view in 1 patient (4%). Ultrasonography revealed the tumors to be dome-shaped in 83%, collar button in 17%, and 8% (n = 2) exhibited intraneural invasion. Mean largest basal dimension was 10.8 mm (SD ± 3.4, median: 11.4, range: 4.6-16.4). Mean tumor thickness was 3.4 mm (SD ± 1.7, median: 2.8, range: 1.4 to 7.1). Visual acuities started at a median 20/25 (range 20/20 to NLP) and decreased to a median 20/40 (range 20/20 to NLP) with a mean follow-up of 25 months (range: 4-60). Radiation optic neuropathy occurred in 50% (n = 12) patients requiring periodic intravitreal bevacizumab to maintain their vision. However, this study has revealed a 100% local control rate for the 21 patients where slotted plaque brachytherapy was used as primary treatment. The one local failure occurred after failure of conventional plaque irradiation. This radiation resistant tumor was removed by enucleation. No patients have developed metastasis during the study.

Conclusions: When used as primary treatment; slotted plaque radiation therapy was found to offer 100% local control of peripapillary, juxtapapillary and circumpapillary choroidal melanomas (with a mean 25 month follow up). Incorporating the optic nerve within the plaque allowed for inclusion of the entire tumor and margin beneath the plaque. This 5-year study suggests that slotted plaque localization over the targeted zone, decreased geographic miss and improved local control.

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2175 Dosimetric Benefit of New Ophthalmic Radiation Plaques for Uveal Melanoma

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Purpose/Objective(s): Uveal melanoma is often treated with radiation plaques with excellent results, albeit with some toxicity to critical structures of the eye. In our institution, COMS plaques have been primarily used. We sought to determine whether the theoretic dosimetry of the Eye Physics plaque (EP917) would reduce radiation exposure to various critical structures of the eye.

Materials/Methods: Randomly selected, 99 consecutive patients with uveal melanoma treated with COMS I-125 radiation plaques were included. Simulation and dosimetry for these plaques were calculated by the Eye Physics Plaque Simulator. These same 99 treatment plans were re-simulated with an updated version of the software. Then, for each patient, theoretic simulation was run for the new EP917, and the treatment plan dosimetry was compared to the COMS plaque. Each new plan involved loading radiation sources to achieve a similar total therapeutic dose of 85 Gray (Gy) to the tumor apex. Total doses to critical structures of the eyes were then compared head to head between the original simulation and the new theoretical simulation. The differences in measures were described using means, standard deviations, and compared using t-tests. The dose differences between EP917 versus COMS in respect to distances of the tumors from critical structures of the eye were calculated with Pearson correlations between the distance measures and differences in doses, calculated with 95% confidence intervals.

Results: The means for total number of seeds and total dose (Gy) were less by 8.07 [$p<0.001$, 95% confidence interval (CI) -9.054 to -7.086] and 22.57 [$p<0.001$, 95% CI -25.83 to -19.3] respectively, in the EP917 plans versus the COMS. For 99 patients, means for total Gy to the eye origin and macula were less by 4.02 [$p<0.001$, 95% CI -4.57 to -3.47] and 26.02 [$p<0.001$, 95% CI -31.36 to -20.68], respectively, in the EP917 plans versus the COMS. For 99 patients, correlation between distance of tumor from the macula and difference in the dose between EP917 and COMS was 0.74 [95% CI 0.63-0.81].

Conclusions: Total radiation doses to the optic disc, opposite retina, lens, eye origin, and macula were all significantly less in the theoretic simulations of the EP917 plaques than the doses received by their COMS counterpart treatment plans. Additionally, as